Area (3D) and Differential equation

(1)

Q1. Let $f: R \to R$ defined by

$$f(x) = \begin{cases} |x-[x]| ; & \text{when } [x] \text{ is odd} \\ |x-[x]-1| ; & \text{when } [x] \text{ is even} \end{cases}$$

when $[\cdot]$ is GIF. what will be the area bounded by the curve $y = f(x)$, x-axis within $x \in [0, 4]$?

Q2. Area determined by two branches of curve $(y-x)^2 = x^3$ and straight line $x = 1$ is $\frac{1}{d+1}$. then evaluate d.

Q3. Two circles of radii 'a' and 'b' touching externally are inscribed in the area bounded by $y = \sqrt{1-x^2}$ and x-axis. If $b = \frac{1}{2}$ and $a = \frac{1}{k}$ then find K.

Q4. Find eight times the area of the region bounded by the point $P(x,y)$ satisfying $\log_x |\log_y x| > 0$ and $\frac{1}{2} < x < 2$.

Q5. A point $P(x,y)$ moves in xy plane in such a way that $\sqrt{2} \le |x+y| + |x-y| \le 2\sqrt{2}$. Find the area of the region representing all possible positions of the point 'P'.

Q6. A point 'P' moves in xy plane in such a way that $[x+y+1] = [x]$ where $[\cdot]$ is GIF and $x \in (0, 2)$. Find the area of the region representing all positions of the point P.

Q7

Column I	Column II
(a) Area between $y = \lvert\cos^{-1}(\sin x)\rvert - \lvert\sin^{-1}(\cos x)\rvert$ and x-axis where $x \in [\frac{3\pi}{2}, 2\pi]$ is $\frac{k\pi^2}{4}$ then k is	(p) 1

(b) The difference between the radii of the smallest and the largest circles which have their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$ and passes through the point $(2, 2)$ is (a) 6

(c) If $f(0) = 1$, $f(2) = 3$ and $f'(2) = 5$, then $\int_0^1 x f''(2x)\,dx$ is equal to (b) 2

(d) If $f(x) = x^3 + 4x^2 + 6x$, then $f^{-1}(-4)$ is (c) -2.

Q5. P: Consider the region R which is enclosed by the curve $y \geq \sqrt{1-x^2}$ and more $\{|x|, |y|\} \leq 1$. The slope of a family of lines defined as $m(t) = -\sin^2 t + \sin t + 1$, where $(t, 2t + 0.4)$ lies inside the region R.

Q6. Then area of region R is

(a) $8 + \frac{\pi}{2}$ (b) $8 - \frac{\pi}{2}$ (c) $4 + \frac{\pi}{2}$ (d) $4 - \frac{\pi}{2}$.

Q7. All possible values of t are represented by the interval

(a) $[0.14, 1.0]$ (b) $[0.14, 0.5]$

(c) $[0.28, 0.8]$ (d) $[0.28, 1.0]$

Q8. The maximum slope of any member of family is for t is equal to

(a) $\frac{\pi}{6}$ (b) 1 (c) 0.28 (d) NOT.

Q9. A: The area of the region $\{(x,y) : x^2+y^2 \leq 5, ||x|-|y|| \geq 1\}$ is $5\left(\pi - \tan^{-1}\frac{24}{7}\right) - 3$.

R: $4\tan^{-1}\frac{1}{2} = \pi - \tan^{-1}\frac{24}{7}$.

Q10. Find a continuous function $f(x)$ satisfying $x^4 - 4x^2 \leq f(x) \leq 2x^2 - x^3$ for all $x \in [0,2]$, such that the area bounded by $y = f(x)$, $y = x^4 - 4x^2$, the y-axis and the line $x = t$ $(0 \leq t \leq 2)$ is k times the area bounded by $y = f(x)$

$y = 2x^2 - x^3$, $y = 2x^2 - x^3$, y-axis and the line $x = t$ $(0 \le t \le 2)$.

Q11. Find the value of 'a' $(a > 2)$ for which the reciprocal of the area enclosed between $y = \frac{1}{x^2}$, $y = \frac{1}{4(x-1)}$, $x = 2$ and $x = a$ is 'a' itself and for what values of $b \in (1,2)$, the area of the figure bounded by the lines $x = b$ and $x = 2$ is $1 - \frac{1}{b}$.

Q12. Find the area of the region in the 1st quadrant bounded on the left by the y-axis below by the line $y = \frac{x}{4}$, above left by the curve $y = 1 + \sqrt{x}$ and above right by the curve $y = \frac{2}{\sqrt{x}}$.

Q13. A polynomial function $f(x)$ satisfies the condition $f(x+1) = f(x) + 2x + 1$. Find $f(x)$ if $f(0) = 1$. Find also the equations of the pair of tangents from the origin on the curve $y = f(x)$ and compute the area enclosed by the curve and the pair of tangents.

Q14. Consider the circle $x^2 + (y-1)^2 = 1$ and the parabola $y = -\frac{x^2}{4}$. The common tangents to the two curves constitute a triangle ABC, the point A and B being on the x-axis and C on the y-axis. CA produced touches the parabola at P and CB produced touches the parabola at Q.

(a) Find the equation of the common tangent BC.

(b) Find the area of the portion between the upper arc of the circle and the common tangents QC and PC.

(c) Find the area enclosed by the parabola $y = -\frac{x^2}{4}$, the x-axis and the lines AP and BQ.

Q15. Find the value of 'c' for which the area of the figure bounded by the curve $y = 8x^2 - x^5$, the straight lines $x = 1$ and $x = c$ and the abscissa axis equal to $16/3$.

Q16. A figure is bounded by the curves $y = |\sqrt{2}\sin\frac{\pi x}{4}|$, $y=0$, $x=2$ & $x=4$. At what angles to the positive x-axis straight lines must be drawn through $(4,0)$ so that these lines partition the figure into three parts of the same size and each part is bounded by $x=2$

Q17. Sketch the curve and find the area enclosed by the loop in the curve $4y^2 = 4x^2 - x^3$.

Q18. If A_i is the area bounded by $|x-a_i| + |y| = b_i$, $i \in \mathbb{N}$, where $a_{i+1} = a_i + \frac{3}{2}b_i$ and $b_{i+1} = \frac{b_i}{2}$, $a_1 = 0$, $b_1 = 32$, then

(a) $A_3 = 128$ (b) $A_3 = 256$ (c) $\lim_{n\to\infty} \sum_{i=1}^{n} A_i = \frac{8}{3}(\quad$

(d) $\lim_{n\to\infty} \sum_{i=1}^{n} A_i = \frac{4}{3}(16)$?

Q19. The area bounded by the curve $y = f(x)$, $y = x$ and the lines $x=1$, $x=t$ is $\left(t + \sqrt{1+t^2}\right) - \sqrt{2} - 1$ square units, for all $t>1$. If $f(x)$ satisfying $f(x) > x$ for all $x>1$, then $f(x)$ is equal to

(a) $x+1 + \frac{x}{\sqrt{1+x^2}}$, $x>1$ (b) $x + \frac{x}{\sqrt{1+x^2}}$, $x>1$

(c) $1 + \frac{x}{\sqrt{1+x^2}}$, $x>1$ (d) $\frac{x}{\sqrt{1+x^2}}$, $x>1$.

Q20. The area of the region bounded by the curve $ay^2 = (2a-x)x^5$ is to that of the circle whose radius is 'a', is given by the ratio

(a) 4:5 (b) 5:8 (c) 2:3 (d) 3:2.

Q21. Area bounded by the curve $y^2(2a-x) = x^3$ and the line $x = 2a$ is

(a) $3\pi a^2$ (b) $\frac{3\pi a^2}{2}$ (c) $\frac{3\pi a^2}{4}$ (d) none

Q22. Area of the region enclosed between the curves $x = y^2 - 1$ and $x = |y|\sqrt{1-y^2}$ is

(a) 1 (b) 4/3 (c) 2/3 (d) 2.

Q23. If the tangent to the curve $y = 1 - x^2$ at $x = \alpha$, where $0 < \alpha < 1$, meets the axes at P and Q. If α varies, the minimum value of the area of the triangle OPQ is k times the area bounded by the axes and the part of the curve for which $0 < x < 1$, then k is equal to

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{75}{16}$ (c) $\frac{25}{18}$ (d) $\frac{2}{3}$.

Q24. Area bounded by curves $y = [\cos A + \cos B + \cos C]$, $y = [7 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}]$, where [.] is GIF and A, B, C are angles of a triangle) and curve $|x-4| + |y| = 2$ is

(a) 2 (b) 3 (c) 4 (d) 6.

Q25. The area of the region defined by $||x| - |y|| \le 1$ and $x^2 + y^2 \le 1$ in the x-y co-ordinate plane is

(a) 1 (b) 2 (c) π (d) NOT.

Q26. The area bounded by $e^{\ln(x+1)} \ge |y|, |x| \le 1$ is

(a) 1 (b) 2 (c) 4 (d) NOT.

Q27. The area bounded by the curves $y = \begin{cases} x^{1/\ln x}, & x \neq 1 \\ e, & x = 1 \end{cases}$ and $y = |x - e|$ is

(a) $e^2/2$ (b) e^2 (c) $2e^2$ (d) 1

Q28. If $f(x) = \sin x \ \forall \ x \in [0, \frac{\pi}{2}]$; $f(x) + f(\pi - x) = 2 \ \forall \ x \in (\frac{\pi}{2}, \pi]$ and $f(x) = f(2\pi - x) \ \forall \ x \in (\pi, 2\pi]$, then the area enclosed by $y = f(x)$ and x-axis is

(a) π (b) 2π (c) 2 (d) 4.

Q29. If $f(x) = x - 1$ and $g(x) = |f(|x|) - 2|$, then the area enclosed by $y = g(x)$ and the curve $x^2 - 4y + 8 = 0$ is equal to

(a) $\frac{4}{3}(4\sqrt{2} - 5)$ (b) $\frac{4}{3}(4\sqrt{2} - 3)$ (c) $\frac{8}{3}(4\sqrt{2} - 3)$ (d) $\frac{8}{3}(4\sqrt{3} - 5)$

Q30. Find the area enclosed between $x^2+y^2 \leq 9\pi^2$ and $\sin(x-y) \geq 0$

(a) $\frac{9\pi^3}{4}$ (b) $\frac{9\pi^3}{2}$ (c) $3\pi^3$ (d) π^3.

Q31. The force of resistance encountered by water on a motor boat of mass 'm' going in still water with velocity 'v' is proportional to the velocity v. At $t=0$ when its velocity is v_0, the engine is shut off. Find an expression for the position of motor boat at time t and also the distance travelled by the boat before it comes to rest. Take the proportionality constant as $K>0$.

Q32. A and B are two separate reservoirs of water. The capacity of reservoir A is double the capacity of reservoir B. Both the reservoirs are filled completely with water, their inlets are closed and then the water is released simultaneously from both the reservoirs. The rate of flow of water out of each reservoir at any instant of time is proportional to the quantity of water in the reservoir at that time. One hour after water is released, the quantity of water in the reservoir A is $1\frac{1}{2}$ times the quantity of water in reservoir B. After how many hours do both the reservoirs have the same quantity of water.

Q33. A tank contains 200 litres of brine in which 20 gms of salt dissolved. Brine containing $\frac{1}{4}$ gm of salt/litre runs into the tank at the rate of 2 litre/min. The mixture is kept stirring runs out out at the same rate. Express the concentration of solution in tank in grams as a function

of tome t. whether the limiting value approached by the amount of salt as $t \to \infty$. when was the amount of salt in solution equal to 20 gm?

Q34. A function $y = f(x)$ satisfies $f'(x) + \frac{2f(x)}{x} = \frac{2\sqrt{f(x)}}{\cos 2x}$ with $f(0) = 1$ then find the value of $f(1)$.

Q35. The function $f: R \to R$ is twice differentiable. Also $f\lambda^2 - 2f'\lambda + f'' = 0$ gives two equal values of λ for every x. Find $f(x)$ if $f(0) = 1$, $f'(0) = 2$.

Q36. The function $y(x)$ satisfies the equation $y(x) + 2x \int_0^x \frac{y(u)}{1+u^2}\,du = 3x^2 + 2x + 1$. Show that the substitution $z(x) = \int_0^x \frac{y(u)}{1+u^2}\,du$ converts the equation into a first order linear differential equation for $z(x)$. Solve for $z(x)$. Hence solve the original equation for $y(x)$.

Q37. Solve $x^2 \frac{dy}{dx} + y^2 e^{\frac{x(y-x)}{y}} = 2y(x-y)$.

Q38. Solve: $(2xy + x^2y + \frac{y^3}{3})dx + (x^2+y^2)\,dy = 0$

Q39. $2y \frac{dy}{dx} = e^{\frac{x^2+y^2}{x}} + \frac{x^2+y^2}{x} - 2x$.

Q40. Find the differential equation corresponding to the family of curves $x^2 + y^2 + 2c_1 + 2c_2 y + c_3 = 0$ where c_1, c_2 & c_3 are arbitrary constants.

Q41. Find the area bounded by the curve $y = \int_{\pi/8}^{\sin 2x} (\sin^{-1}\sqrt{t})\,dt + \int_{\pi/8}^{\cos^2 x} (\cos^{-1}\sqrt{t})\,dt$, $(0 \le x \le \pi/2)$ and the curve satisfying the differential equation $y(x+y^3)\,dx = x(y^3 - x)\,dy$ passing through $(4, -2)$.

Q42. solve $\frac{dy}{dx} = \frac{e^x}{2} \cdot \frac{e^{y^2/x}}{y} - \frac{1}{2}\left(\frac{x}{y} - \frac{y}{x}\right)$.

Q43. solve: $\sqrt{y^2 - x^2y^2 + x^3y^2 - x^5y^2}\, dy = \sqrt{x^4y^2 - x^6 + 2x^4y - x^6y^2 - 2x^6y + x^4}\, dx$

Q44. If the area of a triangle formed by tangent at any point on the curve and co-ordinate axis is constant, then form the differential equations of all such curves and hence prove that the curve is either a straight line or a hyperbola.

Q45. Consider a curve $y = f(x)$ in the xy-plane. The curve passes through $(0,0)$ and has the property that a segment of tangent drawn at any point $P(x, f(x))$ and the line $y = 3$ get bisected by the line $x + y = 1$. Determine the equation of the curve.

Q46. Determine the equation of the curve passing through the points $(3a, a)$ $(a > 0)$ in the form $x = f(y)$, which satisfies the differential equation $\frac{a^2}{xy}\frac{dx}{dy} = \frac{x}{y} + \frac{y}{x} - 2$.

Q47. If a differentiable function $f(x)$ satisfies $\frac{x}{y} f\left(\frac{x}{y}\right) = x f(x) - y f(y)$ $\forall\, x, y \in R$ & $f(e) = \frac{1}{e}$ then prove that $f(x) = \frac{\log x}{x}$.

Q48. If $f: R \to [0, \infty)$ be a function satisfying $f(x+y) - f(x-y) = f(x)\,[f(y) - f(-y)]$; $f'(0) = \log a$, $f(0) = 1$ (for all values of a except) then solve the differential equation $\frac{dy}{dx} = \frac{\{\log_a (f(x) f(y))\}^2}{(\log_a f(x) + 2)(\log_a f(y) - 2)}$

Q49. solve the differential equation $\cos^2 x \frac{dy}{dx} - (\tan 2x)\, y = \cos^4 x$, $|x| < \pi/4$ and $y\left(\frac{\pi}{6}\right) = \frac{3\sqrt{3}}{8}$.

Q50. If a projectile of mass m launched with initial velocity v_0 encounters an air resistance proportional to its velocity, the total force $m\frac{d^2\gamma}{dt^2}$ on the projectile satisfies the equation $m\frac{d^2\gamma}{dt^2} = -mg - k\frac{d\gamma}{dt}$. Find out the resolute vector of the projectile at the particular time instant.

Area and differential equation (1)

1. 2
2. 5
3. 4
4. 7
5. 6
6. 2
7. a - p, b - q, c - r, d - s

8. (6-8) :. 6 - b, 7 - c, 8 - a

9. A - incorrect, R - correct

10. $f(x) = \frac{1}{k+1}\{x^4 - kx^3 + 2(k-2)x^2\}$

11. $a = e^2 + 1$, $b = 1 + e^{-2}$

12. 11/3

13. 2/3

14. a) $y = \sqrt{3}x + 3$ b) $(\sqrt{3} - \frac{\pi}{3})$ c) $\sqrt{3}$

15. $c = (17)^{1/3}, -1, (8 \pm \sqrt{22})^{1/3}$

16. $\pi - (\frac{4\sqrt{2}}{3\pi})$

17. 128/15

18. ~~$\frac{8}{3}(8-2)^2$~~ A, C

19. ~~a~~ 19 - a

20. b

21. a

22. ~~a~~ d

23. a

24. b

25. c

26. c

27. b

28. b

29. a

30 - b

31 → $s(t) = \frac{v_0 m}{k}[1 - e^{-\frac{kt}{m}}]$

time = $\frac{mv_0}{k}$

32. $t_0 = \log_{\frac{3}{4}}(\frac{1}{2})$

33. $y = 50 - 30 \cdot e^{-\frac{1}{100}}$

Amount of salt = 50 gms

34. $yx^2 = [x \tan x + \ln(\cos x)]^2$

35. $f(x) = e^{2x}$

36. $y(x) = \frac{(1+x^2)^2 + 2x}{(1+x^2)}$

37. $x(x-y) = y\ln(ce^x - 1)$

38. $yx^2 + \frac{y^3}{3} = c \cdot e^x$

39. $\ln|cx| = -e^{-\frac{x^2+y^2}{x}}$

40. $\frac{d^3y}{dx^3}[1 + (\frac{dy}{dx})^2] = 3\frac{dy}{dx}(\frac{d^2y}{dx^2})^2$

41. $\frac{1}{8}(\frac{3\pi}{16})^4$

42. ~~$cx = e^{-e}$~~ $-(\frac{x^2+y^2}{x})$

$cx = e^{-e}$

43. $y - \log(y+1) = \frac{2}{3}\sqrt{1+x^3} + C$

~~44. $(y-3)$~~

~~45. $(y-3)^2 = 9(1+x$~~

44. proof.

45. $(y-3)^2 = 9(1-x-y)$

46. $x = y + a\left(\dfrac{1+e^{\frac{2y}{a}-k}}{1-e^{\frac{2y}{a}-k}}\right)$

47. $f(x) = \dfrac{\log x}{x}$

48. $\left(\frac{y-2}{x+2}\right) - \frac{1}{2}\ln\left|1+2\left(\frac{y-2}{x+2}\right)\right| = 2\ln|(x+2)| + C$

49. $y = \frac{1}{2}\ln 2x \cdot \cos 2x$

50. $\vec{r} = -\frac{mg}{k}t + \left(r_0 + \frac{mg}{k}\right)\left(-\frac{m}{k}\right)e^{-\frac{k}{m}t} + C$